

Question:

An achromatic prism combination is to be constructed of crown- and flint-glass components For light of wavelengths 6563 \AA, 5893 \AA, and 4861 \AA, the crown glass has relative indices of refraction of 1.5146, 1.5171, and 1.5233, respectively and the relative indices of refraction of the flint glass are 1.6224, 1.6272, 1.6385, respectively. The wavelengths of 6563 \AA, 5893 \AA, and 4861 \AA, correspond to red, yellow, and blue light, respectively. The crown-glass prism to be used has an angle of 10°. What must be the angle of the flint-glass prism?

Answer Choices:

- A. 10 degrees
- B. 7.20 degrees
- C. 4.45 degrees
- D. 3.75 degrees
- E. 2.50 degrees
- F. 5.40 degrees
- G. 9.00 degrees
- H. 8.15 degrees
- I. 6.30 degrees
- J. 11.00 degrees

Reasoning + Answer:

</tool_call> reasoning process here

For an achromatic prism combination , the condition for eliminating dispersion (i.e., making the deviation of red and violet light equal) is that the net angular dispersion must be zero . This means that the angular dispersion of the crown glass prism must be canceled by that of the flint glass prism .

The angular dispersion ($\delta\theta/\delta\lambda$) of a prism is given by :

$$\delta\theta/\delta\lambda = (d\theta/d\lambda) = (dn/d\lambda)A \quad (1)$$

where A is the angle of the prism and n is the refractive index of the material .

The angular deviation θ is given by :

$$\theta = (n - 1)A \quad (\text{for small angles, } \theta \approx (n - 1)A)$$

But for dispersion , the angular dispersion is proportional to the derivative of $(n - 1)$ with respect to wavelength , i.e., the dispersion curve .

So , the condition for achromatism is :

$$\delta\theta/\delta\lambda_{\text{crown}} + \delta\theta/\delta\lambda_{\text{flint}} = 0$$

i.e.,

$$A_{\text{crown}} \left[(dn_{\text{crown}}/d\lambda) \right] + A_{\text{flint}} \left[(dn_{\text{flint}}/d\lambda) \right] = 0$$

But we can use the formula for dispersion :

Dispersion is proportional to $(n - 1)$ and the derivative of $(n - 1)$ with respect to λ .

So , for each wavelength :

$$\delta\theta/\delta\lambda = A \left(dn/d\lambda \right) \quad (1)$$

For crown glass :

$n_{\text{crown}}(6563) : 1.5146$
 $n_{\text{crown}}(5893) : 1.5171$
 $n_{\text{crown}}(4861) : 1.5233$

For flint glass :

$n_{\text{flint}}(6563) : 1.6224$
 $n_{\text{flint}}(5893) : 1.6272$
 $n_{\text{flint}}(4861) : 1.6385$

We compute the dispersion (change in $(n - 1)$ with λ) for each wavelength .

Let's define $\delta n(\lambda) = (n - 1) \text{ at } \lambda - (n - 1) \text{ at } \lambda_0$

We need to compute the dispersion for each wavelength :

1. For red (6563 \AA):

$\delta n(\lambda)$ for crown : $(1.5146 - 1.5146) = 0$ ~ wait , better to compute differences between wavelengths .

We compute the dispersion (change in $n - 1$) between each pair .

We need the angular dispersion due to each glass at each wavelength .

We use the formula :

$$\text{Dispersion} \approx A \left(dn/d\lambda \right)$$

But since we are matching the dispersion , we can compute the change in $(n - 1)$ over wavelength and then use :

$$\delta\theta/\delta\lambda_{\text{total}} = A_{\text{crown}} \left(\delta n_{\text{crown}}/\delta\lambda \right) + A_{\text{flint}} \left(\delta n_{\text{flint}}/\delta\lambda \right) = 0$$

But we need to compute the dispersion at each wavelength and then use the condition that the net dispersion is zero .

Since the prism is achromatic , the net deviation of red and blue light should be equal , so the dispersion must cancel .

So , we compute the dispersion (change in deviation) due to crown and flint glass for each wavelength pair .

We use the formula :

$$\text{Dispersion} \approx A \left(dn/d\lambda \right)$$

We can use the average of the slopes of $(n - 1)$